

- the data field is new, and a fallback is required for backwards compatibility with legacy instances
- the data field is optional, deprecated, or obsolete and therefore is not always present
- an initial value is needed before the application starts
- the value cannot be determined by the application for the instance
- there is a dependency on the attribute value to formulate other data or affect behavior

If a fallback value is not defined for a data field, the fallback value is determined by the following conditions:

- If the data field is nullable then the fallback value SHALL be NULL
- Else the fallback value SHALL be one of the following, depending on type:
 - Boolean: false
 - Analog: 0 or 0.0, depending on range
 - Bitmaps: 0
 - Enumeration: MS
 - Composite:
 - String: empty
 - List: empty
 - Struct: fallback is recursively composited from the fallback of its member fields
 - Derived types: use the fallback value of the base type

These are the options for the Fallback column used for attributes or attribute, command or event data:

Fallback Column	Description
x	a literal value x of the data type, or as defined in Common Literal Values
MS	a manufacturer or implementation specific value

If the fallback value of a data field is specified as manufacturer specific, then there SHALL be no defined fallback value and the application SHALL support a manufacturer specific value that is in the valid range.

7.19. Data Types

Each data field in a cluster specification SHALL have a well-defined data type. Each attribute in a cluster specification SHALL map to a single data type.

The table indicates for each data type whether it defines an analog or discrete value. Values of analog types MAY be added to or subtracted from other values of the same type and are typically used